

The Quiver associated to the category of G -equivariant perverse sheaves on $M_n(\mathbb{C})$

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1 Introduction

Various categories of perverse sheaves often admit interpretations as categories of representations of quiver [GMV96] [Vil94], but explicit descriptions of these quivers are not easy to obtain. The goal of this article is to find the quiver associated with the category of perverse sheaves with coefficients in $\mathbb{k}$ on the variety of $n \times n$ matrices. An important aspect of this paper is that we can work with field of any characteristics. This paper is perverse analogue of the paper [LW19].

2 The Prophecy on $Perv_{G_n}(X_n, \mathbb{k})$

Let $X_n := M_n(\mathbb{C})$ and $G_n := GL_n(\mathbb{C}) \times GL_n(\mathbb{C})$ be the group acting on X_n by the following action

$$(g, h) \cdot x = gxh^{-1}$$

for all $(g, h) \in G_n$ and $x \in X_n$. The orbits of this action are completely classified by the corank of the matrix. This gives a stratification of X_n in terms of corank of matrices i.e.

$$X_n = \bigsqcup_{0 \leq k \leq n} X_n^k$$

where $X_n^k = \{x \in X_n \mid \text{corank}(x) = k\}$. Note, that the closure of any stratum, X_n^k is given by

$$\overline{X_n^k} = \bigsqcup_{l \geq k} X_n^l$$

and the dimension of stratum X_n^k and $\overline{X_n^k}$ is given by

$$\dim X_n^k = n^2 - k^2 = \dim \overline{X_n^k} \tag{1}$$

for all $0 \leq k \leq n$.

Lemma 1: The stabilizer of any matrix $x \in X_n$ is connected.

Proof. The stabilizer of $x \in X_n$ is the subgroup:

$$\text{Stab}(x) = \{(g, h) \in G_n \mid gx = xh\}$$

Suppose the matrix x has corank r . Up to the action of G_n , any matrix of corank r is in the same orbit as the canonical block matrix:

$$x_r = \begin{pmatrix} I_{n-r} & 0 \\ 0 & 0 \end{pmatrix}$$

where I_{n-r} is the identity matrix of size $n-r$. It is enough to show $Stab(x_r)$ is connected.

Let $g, h \in GL_n(\mathbb{C})$ be partitioned into the same block structure

$$g = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \quad h = \begin{pmatrix} P & Q \\ R & S \end{pmatrix}$$

where A and P are $(n-r) \times (n-r)$ blocks. The stabilizer condition $gx_r = x_rh$ implies $A = P, C = 0$ and $Q = 0$.

Since g and h are invertible, their diagonal blocks A, D , and S must also be invertible. Thus, the stabilizer of x_r takes the form:

$$Stab(x_r) = \left\{ \left(\begin{pmatrix} A & B \\ 0 & D \end{pmatrix}, \begin{pmatrix} A & 0 \\ R & S \end{pmatrix} \right) \mid A \in GL_{n-r}(\mathbb{C}), D, S \in GL_r(\mathbb{C}), B, R^T \in M_{n-r,r}(\mathbb{C}) \right\}.$$

The Levi factor is the block diagonal matrices determined by (A, D, S) which forms a subgroup isomorphic to $GL_{n-r}(\mathbb{C}) \times GL_r(\mathbb{C}) \times GL_r(\mathbb{C})$. The unipotent radical are the matrices parameterized by B and R forming a normal subgroup isomorphic to the additive group of complex matrices $M_{n-r,r}(\mathbb{C}) \oplus M_{r,n-r}(\mathbb{C})$. Since the stabilizer is a semidirect product of two connected spaces, the entire stabilizer group $Stab(x_r)$ is connected. $\square$

The local systems are building blocks of perverse sheaves. The following lemma describes the category of local systems on each strata X_n^k .

Lemma 2: For $0 \leq k \leq n$, the category of local systems $Loc_{G_n}(X_n^k)$ is equivalent to the category of finitely generated $\mathbb{k}$ -modules.

Proof. The strata X_n^k 's are homogeneous with respect to G_n , using 1 and proposition 6.2.13 in [Ach21], there are no non trivial local G_n equivariant local system X_n^k . $\square$

Let $\mathbb{k}$ be a field. Let $D_{G_n}^b(X_n, \mathbb{k})$ be the bounded derived category of G_n -equivariant complexes on X_n with coefficient in $\mathbb{k}$. Consider $Perv_{G_n}(X_n, \mathbb{k}) \subset D_{G_n}^b(X_n, \mathbb{k})$ the subcategory of G_n -equivariant perverse sheaves on X_n with coefficients in $\mathbb{k}$.

The above lemma shows that the category of local systems has enough projectives. This information is important for our next theorem.

Theorem 3: The category $Perv_{G_n}(X_n, \mathbb{k})$ has enough projectives.

Proof. The category of G_n -equivariant perverse sheaves can be realized as perverse sheaves constructible with respect to the G_n orbit stratification. Since there are finitely many orbits, using 2 and theorem 4.6 in [CW22] we get our result. $\square$

When there are enough projective perverse sheaves and finitely many simple, the category of perverse sheaves can be described as modules over a finite-dimensional algebra, and therefore also as representations of a quiver with relations. The rest of the story is the quest to find this quiver.

3 Building the Quiver

Let $\text{rep}(\widehat{Q})$ be the category of finite-dimensional representations of quiver with relations $\widehat{Q} = (Q, I)$. The (isomorphism classes of) simple objects in $\text{rep}(\widehat{Q})$ are in bijection with the vertices of Q . For each vertex x , the corresponding simple S^x is the representation with $(S^x)_x = \mathbb{k}$ and $(S^x)_y = 0$ for all other vertex $y \neq x$. Let P^x (resp. I^x) denote the projective cover (resp. injective envelope) of S^x , as constructed in [[ASS06], III.2]. In particular, for vertex y , the dimension of $(P^x)_y$ (resp. $(I^x)_y$) is given by the number of paths from x to y (resp. from y to x), considered up to the relations in I .

We will be using this information to retrieve the quiver from the category of perverse sheaves. The simple objects of the category will be in one-to-one correspondence with the vertices of the quiver. Moreover, the number of edges between vertex x to vertex y will be calculated by finding the dimension of first extension group i.e. $\dim_{\mathbb{k}} \text{Ext}^1(S_x, S_y)$ where S_x and S_y are the simple perverse sheaves corresponding to the vertex x and y respectively.

3.1 Finding the Vertices

The simple objects in $\text{Perv}_{G_n}(X_n, \mathbb{k}) \subset D_{G_n}^b(X_n, \mathbb{k})$ are parametrized by intersection complexes i.e. $IC(X_n^k, L)$ where L is an irreducible G_n equivariant local system of X_n^k . Using 2, there is only unique irreducible local systems i.e. the trivial local system.

Therefore,

$$IC_k := IC(X_n^k, \mathbb{k}_{X_n^k}[\dim X_n^k])$$

where $0 \leq k \leq n$ are all the simple objects in $\text{Perv}_{G_n}(X_n, \mathbb{k}) \subset D_{G_n}^b(X_n, \mathbb{k})$. This also tells us that there are $n + 1$ vertices in the quiver associated to $\text{Perv}_{G_n}(X_n, \mathbb{k})$.

$$0 \quad 1 \quad 2 \quad \dots \quad n$$

The vertex k in the above quiver corresponds to the simple perverse sheaf IC_k .

We will give an alternate description of IC_k 's which will help us to compute their stalks. Let

$$\widetilde{X}_n^k = \{(x, W) \in \overline{X}_n^k \times Gr(k, n) \mid W \subset \ker(x)\}$$

where $Gr(k, n)$ is the Grassmanian variety of k -dimensional spaces in $\mathbb{C}^n$. Let G_n act on $\widetilde{X}_n^k$ by

$$(g_1, g_2) \cdot (x, W) = (g_1 x g_2^{-1}, g_1 \cdot W)$$

for all $(g_1, g_2) \in G_n$. Under the usual $GL_n(\mathbb{C})$ -action on $Gr(k, n)$, the second projection $p_k : \widetilde{X}_n^k \rightarrow Gr(k, n)$ is $GL_n(\mathbb{C})$ -equivariant and the fibers are of the same dimension. This makes p_k a vector bundle over a smooth variety. Thus, $\widetilde{X}_n^k$ is a smooth variety for all k .

We stratify $\widetilde{X}_n^k$ by using the stratification on $\overline{X}_n^k$ i.e.

$$\widetilde{X}_n^k = \bigsqcup_{l \geq k} \pi_k^{-1}(X_n^l).$$

Lemma 4: The first projection map $\pi_k : \widetilde{X}_n^k \rightarrow \overline{X}_n^k$ is proper and small with respect to X_n^k .

Proof. Consider the first projection map $\Pi_k : \overline{X}_n^k \times Gr(k, n) \rightarrow \overline{X}_n^k$. Since $Gr(k, n)$ is a projective variety, Π_k is proper. Moreover, $\widetilde{X}_n^k$ is a closed subscheme of $\overline{X}_n^k \times Gr(k, n)$ which implies the restriction $\pi_k = \Pi_k|_{\widetilde{X}_n^k}$ is proper.

The biggest strata of $\widetilde{X}_n^k$ is $\pi_k^{-1}(X_n^k) = X_n^k \times Gr(k, k) \cong X_n^k$. Using (1) we get

$$\dim \widetilde{X}_n^k = n^2 - k^2.$$

For $l \geq k$ and $x_n^l \in X_n^l$, the fiber over x_n^l i.e. $\pi_k^{-1}(x_n^l)$ is isomorphic to $Gr(k, l)$ which gives

$$\dim \pi_k^{-1}(x_n^l) = \dim Gr(k, l) = k(l - k).$$

Note that X_n^k is an open, dense set in $\overline{X}_n^k$ and for $x_n^l \in X_n^l$ where $l > k$,

$$\dim \pi_k^{-1}(x_n^l) = k(l - k) < \frac{1}{2}(l^2 - k^2) = \frac{1}{2}(\dim \widetilde{X}_n^k - \dim X_n^l).$$

Thus, the map $\pi_k : \widetilde{X}_n^k \rightarrow \overline{X}_n^k$ is small with respect to X_n^k . $\square$

Corollary 5: For $0 \leq k \leq n$,

$$IC_k \cong \pi_{k*} \mathbb{K}_{\widetilde{X}_n^k}[\dim X_n^k]. \quad (2)$$

Proof. Since the maps π_k are small with respect to X_n^k and $\pi_k^{-1}(X_n^k) = X_n^k \times Gr(k, k) \cong X_n^k$, we get our result using (see 3.8.10 in [Ach21]). $\square$

It becomes easier to calculate stalks with the above description of IC objects. The following theorem gives a local description of our simple objects.

Theorem 6: For $n \geq l \geq k \geq 0$ and $t > -(l - k)^2$,

$$\begin{aligned} {}^p H^t(IC_k|_{X_n^l}, \mathbb{K}) &= 0 \\ {}^p H^t(IC_k|_{\overline{X}_n^l}, \mathbb{K}) &= 0. \end{aligned}$$

Proof. Using 2, for $x_n^l \in X_n^l$,

$$H^t((\pi_{k*} \mathbb{K}_{\widetilde{X}_n^k}[\dim X_n^k])_{x_n^l}, \mathbb{K}) = H_{singular}^{t+\dim X_n^k}(\pi_k^{-1}(x_n^l), \mathbb{K}) = H_{singular}^{t+\dim X_n^k}(Gr(k, l), \mathbb{K})$$

for all $k \leq l \leq n$. The top cohomology of $Gr(k, l)$ lies in degree $2k(l - k)$ which implies

$$H^t(\pi_{k*} \mathbb{K}_{\widetilde{X}_n^k}[\dim X_n^k]_{X_n^l}, \mathbb{K}) = 0$$

for $t > 2k(l - k) - \dim X_n^k = -\dim X_n^l - (l - k)^2$. This implies

$${}^p H^t(\pi_{k*} \mathbb{K}_{\widetilde{X}_n^k}[\dim X_n^k]_{X_n^l}, \mathbb{K}) = H^{t-\dim X_n^l}(\pi_{k*} \mathbb{K}_{\widetilde{X}_n^k}[\dim X_n^k]_{X_n^l}, \mathbb{K}) = 0$$

for $t > -(l - k)^2$.

Since $\overline{X}_n^l = \sqcup_{i \geq l} X_n^i$, and for $i \geq l$,

$${}^p H^t(\pi_{k*} \mathbb{K}_{\widetilde{X}_n^k}[\dim X_n^k]_{X_n^i}, \mathbb{K}) = 0$$

for all $t > -(l - k)^2 \geq -(i - k)^2$, we get ${}^p H^t(\pi_{k*} \mathbb{K}_{\widetilde{X}_n^k}[\dim X_n^k]_{\overline{X}_n^l}, \mathbb{K}) = 0$ for $t > -(l - k)^2$ from Lemma 3.1.6 [Ach21]. $\square$

In particular, for the inclusion map. $i_{l,k} : \overline{X}_n^l \hookrightarrow \overline{X}_n^k$

$$i_{l,k}^* IC_k \in {}^p D_{G_n}^b(\overline{X}_n^l, \mathbb{k})^{\leq -(l-k)^2}. \quad (3)$$

Since IC_k 's are invariant under Verdier duality, we get

$$i_{l,k}^! IC_k \cong \mathbb{D}(i_{l,k}^* \pi_{k*} \mathbb{k}_{\widetilde{X}_n^k}[\dim X_n^k]) \in {}^p D_{G_n}^b(\overline{X}_n^l, \mathbb{k})^{\geq (l-k)^2}. \quad (4)$$

3.2 Forging the Edges

In this section, we will use the above results and calculate $Ext^1(IC_k, IC_l)$ for $0 \leq k, l \leq n$.

Lemma 7: The first G_n equivariant cohomology of X_n vanishes i.e.

$$H_{G_n}^1(X_n, \mathbb{k}) = Hom(\mathbb{k}_{X_n}, \mathbb{k}_{X_n}[1]) = 0.$$

Proof. Observe that X_n retracts G_n -equivariantly to the matrix $\{0_n\}$ via homotopy $t \mapsto tx$. Thus their equivariant cohomologies are isomorphic i.e

$$H_{G_n}^1(X_n, \mathbb{k}) \cong H_{G_n}^1(\{0_n\}, \mathbb{k}) \cong H^1(BG_n, \mathbb{k})$$

where BG_n is the classifying space of G_n . Since $G_n = GL_n(\mathbb{C}) \times GL_n(\mathbb{C})$, we get $BG_n \simeq BGL_n(\mathbb{C}) \times BGL_n(\mathbb{C})$. The theorem 14.5 in [MS74] shows that cohomology ring of $BGL_n(\mathbb{C})$ is a polynomial ring generated by the universal Chern classes

$$H^*(BGL_n(\mathbb{C}), \mathbb{k}) \cong \mathbb{k}[c_1, c_2, \dots, c_n],$$

where the topological degree of each Chern class is strictly even (i.e. $\deg(c_i) = 2i$). By the Künneth formula, $H^*(BG_n, \mathbb{k})$ is the tensor product of two such polynomial rings. Since all generators exist in strictly even degrees, the cohomology groups in all odd degrees must vanish. Therefore, $H_G^1(X_n, \mathbb{k}) = 0$. $\square$

In a similar fashion, the above lemma is true replacing X_n with the orbit X_n^k .

Lemma 8: The first G_n equivariant cohomology of X_n^k vanishes i.e.

$$H_{G_n}^1(X_n^k, \mathbb{k}) = Hom(\mathbb{k}_{X_n^k}, \mathbb{k}_{X_n^k}[1]) = 0.$$

Proof. Using the orbit-stabilizer theorem, we have

$$X_n^k \cong G_n / Stab(x_k).$$

Moreover,

$$H_{G_n}^1(X_n^k, \mathbb{k}) \cong H_{G_n}^1(G_n / Stab(x_k), \mathbb{k}) \cong H^1(BStab(x_k), \mathbb{k})$$

where $BStab(x_k)$ is the classifying space of $Stab(x_k)$.

From 2, the unipotent radical of $Stab(x_k)$ i.e. $M_{n-r,r}(\mathbb{C}) \oplus M_{r,n-r}(\mathbb{C})$ is contractible. So, $Stab(x_k)$ retracts to its Levi subgroup i.e. $GL_{n-r} \times GL_r \times GL_r$. Thus,

$$H^*(BStab(x_k), \mathbb{k}) \cong H^*(BGL_{n-r}(\mathbb{C}), \mathbb{k}) \otimes H^*(BGL_r(\mathbb{C}), \mathbb{k}) \otimes H^*(BGL_r(\mathbb{C}), \mathbb{k})$$

by the Künneth formula. The theorem 14.5 in [MS74] shows that cohomology ring of $BGL_n(\mathbb{C})$ is a polynomial ring generated by the universal Chern classes

$$H^*(BGL_n(\mathbb{C}), \mathbb{k}) \cong \mathbb{k}[c_1, c_2, \dots, c_n],$$

where the topological degree of each Chern class is strictly even (i.e. $\deg(c_i) = 2i$). Since all generators exist in strictly even degrees, the cohomology groups in all odd degrees must vanish. Therefore, $H_G^1(X_n^k, \mathbb{k}) = 0$. $\square$

The following theorem will tell us that there are no loops in our associated quiver.

Theorem 9: For $0 \leq k \leq n$,

$$Ext^1(IC_k, IC_k) = 0$$

Proof. Consider the distinguished triangle,

$$j_{k!}\mathbb{k}_{X_n^k}[\dim X_n^k] \rightarrow IC_k \rightarrow i_{k!}i_k^*IC_k \rightarrow,$$

where $j_k : X_n^k \hookrightarrow \overline{X_n^k}$ is open embedding and i_k is complementary closed embedding. Applying the functor $Hom(-, IC_k[1])$ and adjointness relation, we get the following long exact sequence

$$Hom(\mathbb{k}_{X_n^k}[\dim X_n^k], \mathbb{k}_{X_n^k}[\dim X_n^k + 1]) \leftarrow Ext^1(IC_k, IC_k) \leftarrow Hom(i_k^*IC_k, i_k^!IC_k[1]) \leftarrow .$$

Using 8, we have

$$Hom(\mathbb{k}_{X_n^k}[\dim X_n^k], \mathbb{k}_{X_n^k}[\dim X_n^k + 1]) = 0.$$

From 3 and 4, we can see that $i_k^*IC_k \in {}^pD_{G_n}^b(\overline{X_n^l}, \mathbb{k})^{\leq -1}$ and $i_k^!IC_k[1] \in {}^pD_{G_n}^b(\overline{X_n^l}, \mathbb{k})^{\geq 0}$ which implies

$$Hom(i_k^*IC_k, i_k^!IC_k[1]) = 0.$$

Thus,

$$Ext^1(IC_k, IC_k) = 0$$

for all $0 \leq k \leq n$. $\square$

The following theorem tells us that there are no edges between non consecutive vertices.

Theorem 10: For $0 \leq k, l \leq n$, and $l - k \geq 2$,

$$Ext^1(IC_k, IC_l) = 0$$

$$Ext^1(IC_l, IC_k) = 0$$

Proof. Let $i_{l,k} : \overline{X_n^l} \hookrightarrow \overline{X_n^k}$. Since IC_l is supported on $\overline{X_n^l} \subset \overline{X_n^k}$,

$$\begin{aligned} Ext^1(IC_k, IC_l) &= Hom(IC_k, i_{l,k*}IC_l[1]) \\ &= Hom(i_{l,k}^*IC_k, IC_l[1]) \\ &= 0 \end{aligned} \quad (by \ 3)$$

and

$$\begin{aligned}
Ext^1(IC_l, IC_k) &= Hom(i_{l,k*}IC_l, IC_k[1]) \\
&= Hom(i_{l,k}!IC_l, IC_k[1]) \\
&= Hom(IC_l, i_{l,k}^!IC_k[1]) \\
&= 0.
\end{aligned} \tag{by 4}$$

Hence, for $0 \leq k, l \leq n$, and $l - k \geq 2$,

$$\begin{aligned}
Ext^1(IC_k, IC_l) &= 0 \\
Ext^1(IC_l, IC_k) &= 0.
\end{aligned}$$

□

Let

$$j_k : X_n^k \hookrightarrow X_n$$

be the inclusion map of stratum X_n^k in X_n . We will construct the projective cover and injective hull of IC_0 and IC_n .

Lemma 11: Let $U_k = X_n^k \cup X_n^{k+1}$ and $f_k : U_k \rightarrow \overline{X_n^k}$ be the open embedding. Then,

$$\begin{aligned}
{}^pH^0(f_{k*}(IC_k|_{U_k})) &= IC_k \\
{}^pH^1(f_{k*}(IC_k|_{U_k})) &= 0
\end{aligned}$$

Proof. The map $i_{k+2,k} : \overline{X_n^{k+2}} \hookrightarrow \overline{X_n^k}$ is the complementary closed embedding to f_k . We have a distinguished triangle

$$i_{k+2,k*}i_{k+2,k}^!IC_k \rightarrow IC_k \rightarrow f_{k*}(IC_k|_{U_k}) \rightarrow .$$

Since IC_k is perverse, only the zeroth perverse cohomology is non zero and

$${}^pH^0(IC_k) = IC_k.$$

Using (4) and $i_{k+2,k*}$ is t-exact, we get

$${}^pH^t(i_{k+2,k*}i_{k+2,k}^!IC_k) = 0$$

for $t \leq 3$. Taking the long exact sequence of perverse cohomology of the above distinguished triangle, we get the desired isomorphisms. □

Theorem 12: Let $0 \leq k \leq n-1$, $U_k := X_n^k \cup X_n^{k+1}$ and $h_k : X_n^{k+1} \hookrightarrow U_k$ be the closed embedding and $j_k : X_n^k \hookrightarrow \overline{X_n^k}$ be the open embedding.

$$\begin{aligned}
{}^pH^0(j_{k+1*}h_k^!(IC_k|_{U_k})) &= 0 \\
{}^pH^1(j_{k+1*}h_k^!(IC_k|_{U_k})) &= {}^pH^0(j_{k+1*}\underline{\mathbb{K}}_{X_n^{k+1}}[\dim X_n^{k+1}])
\end{aligned}$$

Proof. From 4, we have $h_k^!(IC_k|_{U_k}) \in {}^pD^{\geq 1}$ and j_{k+1*} is left t-exact which implies $j_{k+1*}h_k^!(IC_k|_{U_k}) \in {}^pD^{\geq 1}$. Thus, ${}^pH^0(j_{k+1*}h_k^!(IC_k|_{U_k})) = 0$.

Using proper base change theorem, we can write

$$IC_k|_{U_k} \cong \pi_{k*} \mathbb{K}_{\widetilde{X_n^k}}[\dim X_n^k]|_{U_k} \cong \pi'_{k*} \mathbb{K}_{\pi_k^{-1}(U_k)}[\dim X_n^k]$$

where $\pi'_k : \pi_k^{-1}(U_k) \xrightarrow{\pi_k} U_k$.

The term $h_k^! \pi_{k*} \mathbb{K}_{\widetilde{X_n^k}}[\dim X_n^k]|_{U_k} \cong h_k^! \pi'_{k*} \mathbb{K}_{\pi_k^{-1}(U_k)}[\dim X_n^k]$ can be further simplified using the following Cartesian square

$$\begin{array}{ccc} \pi_k^{-1}(X_n^{k+1}) & \xrightarrow{h'_k} & \pi_k^{-1}(U_k) \\ \pi''_k \downarrow & & \downarrow \pi'_k \\ X_n^{k+1} & \xrightarrow{h_k} & U_k \end{array}$$

to get

$$h_k^! \pi'_{k*} \mathbb{K}_{\pi_k^{-1}(U_k)}[\dim X_n^k] \cong \pi''_{k*} h_k^! \mathbb{K}_{\pi_k^{-1}(U_k)}[\dim X_n^k].$$

Since $\pi_k^{-1}(U_k)$ is smooth and $\dim(\pi_k^{-1}(U_k)) = \dim X_n^k$, we can write

$$\begin{aligned} h_k^! \mathbb{K}_{\pi_k^{-1}(U_k)}[\dim X_n^k] &\cong h_k^! w_{\pi_k^{-1}(U_k)}[-\dim X_n^k] \\ &\cong w_{\pi_k^{-1}(X_n^{k+1})}[-\dim X_n^k] \\ &\cong \mathbb{K}_{\pi_k^{-1}(X_n^{k+1})}[2\dim(\pi_k^{-1}(X_n^{k+1})) - \dim X_n^k] \\ &\cong \mathbb{K}_{\pi_k^{-1}(X_n^{k+1})}[2(\dim X_{k+1}^n + k) - \dim X_n^k] \\ &\cong \mathbb{K}_{\pi_k^{-1}(X_n^{k+1})}[\dim X_{k+1}^n + 2k + \dim X_{k+1}^n - \dim X_n^k] \\ &\cong \mathbb{K}_{\pi_k^{-1}(X_n^{k+1})}[\dim X_{k+1}^n - 1] \end{aligned}$$

Since the fibers of π''_k are isomorphic to $\mathbb{P}^k$ and j_{k+1*} is left t-exact, we have the following

$$\begin{aligned} {}^pH^1(j_{k+1*}h_k^!(IC_k|_{U_k})) &\cong {}^pH^1(j_{k+1*}\pi''_{k*}h_k^! \mathbb{K}_{\pi_k^{-1}(U_k)}[\dim X_n^k]) \\ &\cong {}^pH^1(j_{k+1*}\pi''_{k*} \mathbb{K}_{\pi_k^{-1}(X_n^{k+1})}[\dim X_n^{k+1} - 1]) \\ &\cong {}^pH^0(j_{k+1*}\pi''_{k*} \mathbb{K}_{\pi_k^{-1}(X_n^{k+1})}[\dim X_n^{k+1}]) \\ &\cong {}^p\tau^{\leq 0}(j_{k+1*}\pi''_{k*} \mathbb{K}_{\pi_k^{-1}(X_n^{k+1})}[\dim X_n^{k+1}]) \\ &\cong {}^p\tau^{\leq 0}(j_{k+1*} {}^pH^0(\pi''_{k*} \mathbb{K}_{\pi_k^{-1}(X_n^{k+1})}[\dim X_n^{k+1}])) \\ &\cong {}^p\tau^{\leq 0}(j_{k+1*} \mathbb{K}_{X_n^{k+1}}) \\ &\cong {}^p\tau^{\leq 0}(j_{k+1*} {}^pH^0(\mathbb{K}_{X_n^{k+1}}[\dim X_n^{k+1}])) \\ &\cong {}^p\tau^{\leq 0}j_{k+1*}(\mathbb{K}_{X_n^{k+1}}[\dim X_n^{k+1}]) \\ &\cong {}^pH^0(j_{k+1*} \mathbb{K}_{X_n^{k+1}}[\dim X_n^{k+1}]) \end{aligned}$$

□

The following theorem will describe composition series of some important perverse objects.

Theorem 13: Let $0 \leq k \leq n-1$ and $j_k : X_n^k \hookrightarrow \overline{X_n^k}$ be the inclusion map. Then, we get following exact sequence

$$0 \rightarrow IC_k \rightarrow {}^p H^0(j_{k*} \underline{\mathbb{K}}_{X_n^k}[\dim X_n^k]) \rightarrow {}^p H^0(j_{k+1*} \underline{\mathbb{K}}_{X_n^{k+1}}[\dim X_n^{k+1}]) \rightarrow 0$$

and

$$0 \rightarrow {}^p H^0(j_{k+1!} \underline{\mathbb{K}}_{X_n^{k+1}}[\dim X_n^{k+1}]) \rightarrow {}^p H^0(j_{k!} \underline{\mathbb{K}}_{X_n^k}[\dim X_n^k]) \rightarrow IC_k \rightarrow 0$$

Proof: Let $U_k := X_n^k \cup X_n^{k+1}$ and consider the open embedding $g_k : X_n^k \hookrightarrow U_k$ and the complementary closed embedding $h_k : X_n^{k+1} \hookrightarrow U_k$. We get a distinguished triangle

$$h_{k*} h_k^!(IC_k|_{U_k}) \rightarrow IC_k|_{U_k} \rightarrow g_{k*} \underline{\mathbb{K}}_{X_n^k}[\dim X_n^k] \rightarrow .$$

Let $f_k : U_k \hookrightarrow \overline{X_n^k}$ be the inclusion map. Applying the pushforward f_{k*} to the above distinguished triangle we get

$$j_{k+1*} h_k^!(IC_k|_{U_k}) \rightarrow f_{k*} IC_k|_{U_k} \rightarrow j_{k*} \underline{\mathbb{K}}_{X_n^k}[\dim X_n^k] \rightarrow .$$

Taking perverse cohomology, we get the following long exact sequence

$$\begin{aligned} {}^p H^0(j_{k+1*} h_k^!(IC_k|_{U_k})) &\rightarrow {}^p H^0(f_{k*} IC_k|_{U_k}) \rightarrow {}^p H^0(j_{k*} \underline{\mathbb{K}}_{X_n^k}[\dim X_n^k]) \\ &\rightarrow {}^p H^1(j_{k+1*} h_k^!(IC_k|_{U_k})) \rightarrow {}^p H^1(f_{k*} IC_k|_{U_k}) \rightarrow \end{aligned}$$

Using 11 and 12, we get the desired result. Applying the Verdier dual we get the other short exact sequence.

Corollary 14: For $0 \leq k \leq n$, the perverse sheaf ${}^p H^0(j_{k*} \underline{\mathbb{K}}_{X_n^k}[\dim X_n^k])$ (respectively ${}^p H^0(j_{k!} \underline{\mathbb{K}}_{X_n^k}[\dim X_n^k])$) is uniserial with composition factors $IC_k, \dots, IC_n$ (respectively $IC_n, \dots, IC_k$).

Lemma 15: The perverse sheaf ${}^p H^0(j_{0!} \underline{\mathbb{K}}_{X_n^0}[\dim X_n^0])$ (respectively ${}^p H^0(j_{0*} \underline{\mathbb{K}}_{X_n^0}[\dim X_n^0])$) is the projective cover (respectively the injective envelope) of IC_0 .

Proof. Since j_0 is an open embedding, it is t -exact and we have

$$\begin{aligned} Hom({}^p H^0(j_{0!} \underline{\mathbb{K}}_{X_n^0}[\dim X_n^0]), {}^p H^0(j_{0!} \underline{\mathbb{K}}_{X_n^0}[\dim X_n^0])) &= Hom({}^p \tau^{\geq 0}(j_{0!} \underline{\mathbb{K}}_{X_n^0}[\dim X_n^0]), {}^p H^0 \geq 0(j_{0!} \underline{\mathbb{K}}_{X_n^0}[\dim X_n^0])) \\ &= Hom(j_{0!} \underline{\mathbb{K}}_{X_n^0}[\dim X_n^0], {}^p H^0 j_{0!} \underline{\mathbb{K}}_{X_n^0}[\dim X_n^0]) \\ &= Hom(\underline{\mathbb{K}}_{X_n^0}[\dim X_n^0], j_0^* {}^p H^0 j_{0!} \underline{\mathbb{K}}_{X_n^0}[\dim X_n^0]) \\ &= Hom(\underline{\mathbb{K}}_{X_n^0}[\dim X_n^0], {}^p H^0 j_0^* j_{0!} \underline{\mathbb{K}}_{X_n^0}[\dim X_n^0]) \\ &= Hom(\underline{\mathbb{K}}_{X_n^0}[\dim X_n^0], \underline{\mathbb{K}}_{X_n^0}[\dim X_n^0]) \cong \mathbb{K}. \end{aligned}$$

Thus, the objects ${}^p H^0(j_{0!} \underline{\mathbb{K}}_{X_n^0}[\dim X_n^0])$ is indecomposable.

Moreover,

$$\begin{aligned} Hom((j_{0!} \underline{\mathbb{K}}_{X_n^0}[\dim X_n^0]), IC_k[1]) &= Hom(\underline{\mathbb{K}}_{X_n^0}[\dim X_n^0], j_0^! IC_k[1]) \\ &= Hom(\underline{\mathbb{K}}_{X_n^0}[\dim X_n^0], j_0^* IC_k[1]) \end{aligned}$$

and

$$j_0^* IC_k[1] = \begin{cases} \mathbb{K}_{X_n^0}[\dim X_n^0 + 1], & \text{if } k = 0 \\ 0, & \text{otherwise} \end{cases},$$

using 8, we get $Hom(j_0! \mathbb{K}_{X_n^0}[\dim X_n^0], IC_k[1]) = 0$ for all $0 \leq k \leq n$.

Consider the distinguished triangle

$$j_0! \mathbb{K}_{X_n^0}[\dim X_n^0] \rightarrow {}^p H^0(j_0! \mathbb{K}_{X_n^0}[\dim X_n^0]) \rightarrow {}^p \tau^{<0} j_0! \mathbb{K}_{X_n^0}[\dim X_n^0][1] \rightarrow$$

Applying $Hom(-, IC_k[1])$ functor to the distinguished triangle we get

$$Hom({}^p \tau^{<0} j_0! \mathbb{K}_{X_n^0}[\dim X_n^0], IC_k) \rightarrow Ext^1({}^p H^0(j_0! \mathbb{K}_{X_n^0}[\dim X_n^0]), IC_k) \rightarrow Hom(j_0! \mathbb{K}_{X_n^0}[\dim X_n^0], IC_k[1])$$

Since ${}^p \tau^{<0} j_0! \mathbb{K}_{X_n^0}[\dim X_n^0] \in {}^p D^{<0}$ and $IC_k \in {}^p D^{\geq 0}$, we get

$$Hom({}^p \tau^{<0} j_0! \mathbb{K}_{X_n^0}[\dim X_n^0], IC_k) = 0.$$

Hence, $Hom(j_0! \mathbb{K}_{X_n^0}[\dim X_n^0], IC_k[1]) = 0$ implies

$$Ext^1({}^p H^0(j_0! \mathbb{K}_{X_n^0}[\dim X_n^0]), IC_k).$$

Using 13, we see ${}^p H^0(j_0! \mathbb{K}_{X_n^0}[\dim X_n^0])$ has a unique simple subobject i.e. IC_0 . Hence, it is the projective cover of IC_0 . Using Verdier duality, $\mathbb{D}({}^p H^0(j_0! \mathbb{K}_{X_n^0}[\dim X_n^0])) \cong {}^p H^0(j_0* \mathbb{K}_{X_n^0}[\dim X_n^0])$ we get injective envelope of IC_0 . $\square$

We can also construct injective envelope and projective cover of IC_n .

Corollary 16: Let $\mathbb{T}$ be the Fourier-Deligne transform functor. Then the perverse sheaf $\mathbb{T}({}^p H^0(j_0! \mathbb{K}_{X_n^0}[\dim X_n^0]))$ (respectively $\mathbb{T}({}^p H^0(j_0* \mathbb{K}[\dim X_n^0]))$) is the projective cover (respectively the injective envelope) of IC_n .

Proof. The functor $\mathbb{T}$ is an equivalence of categories that preserves projectives, injectives and simples. It also takes the constant sheaf IC_0 to the sky scrapper sheaf IC_n . Since ${}^p H^0(j_0! \mathbb{K}_{X_n^0}[\dim X_n^0])$ is the projective cover of IC_0 and ${}^p H^0(j_0* \mathbb{K}_{X_n^0}[\dim X_n^0])$ is the injective envelope of IC_n , we get our result. $\square$

Remark 17: Using 14 and 16, the multiplicity of IC_k in the projective cover of IC_0 and the projective cover of IC_n is one.

Using 15 and 14, the composition series of projective cover of IC_0 gives us the following path in the associated quiver:

$$0 \xrightarrow{u_0} 1 \xrightarrow{u_1} 2 \xrightarrow{u_2} \dots \xrightarrow{u_{n-1}} n.$$

Note that there is only one edge between consecutive vertices because the multiplicity of IC_k 's in the projective cover of IC_0 i.e. ${}^p H^0(j_0! \mathbb{K}_{X_n^0}[\dim X_n^0])$ is one. By Verdier duality, we must have edge in opposite direction as well, i.e.

$$0 \xleftarrow[v_0]{u_0} 1 \xleftarrow[v_1]{u_1} 2 \xleftarrow[v_2]{u_2} \dots \xleftarrow[v_{n-1}]{u_{n-1}} n$$

From 9 and 10, there are no loops and no edges between non consecutive edges. Thus, atleast on level of quivers, we get all the edges.

4 Relations in Quivers

Let Q be a quiver and $\mathbb{k}Q$ be the path algebra with basis given by directed paths in Q and multiplication induced by concatenation of paths.

Definition 18: A *relation* in Q is a $\mathbb{k}$ -linear combination of paths of length at least two having the same source and target. A *quiver with relations* $\widehat{Q} := (Q, I)$ is a quiver with finite set of relations I .

The quiver algebra of $\widehat{Q}$ is the quotient $\mathbb{k}Q / \langle I \rangle$ of the path algebra by the ideal generated by the relations. We will assume that the ideal of relations $\langle I \rangle$ contains any path whose length is large enough, so that the corresponding quiver algebra is finite dimensional.

Example 19: The quiver $\widehat{AA}_n$ is the quiver with $n+1$ vertices with edges and relations are as follows,

$$0 \begin{array}{c} \xrightarrow{u_0} \\ \xleftarrow{v_0} \end{array} 1 \begin{array}{c} \xrightarrow{u_1} \\ \xleftarrow{v_1} \end{array} 2 \begin{array}{c} \xrightarrow{u_2} \\ \xleftarrow{v_2} \end{array} \cdots \begin{array}{c} \xrightarrow{u_{n-1}} \\ \xleftarrow{v_{n-1}} \end{array} n$$

with the relations

$$u_i v_i = 0 = v_i u_i.$$

In [LW19], András C. Lörincz, Uli Walther showed the category of G_n equivariant D -modules on X_n is equivalent to the category of representations of $\widehat{AA}_n$.

We have the quiver as described in the above section but we should also describe relations between edges u_i 's and v_i 's. On the level of quiver, it matches with $\widehat{AA}_n$. We will show that it also has the same relations.

5 Perverse sheaves on $M_n \mathbb{C}$ and representations on $\widehat{AA}_n$

Let $Perv_{G_n}(X_n, \mathbb{k})_n$ be the full subcategory of $Perv_{G_n}(X_n, \mathbb{k})$ consisting of G_n -equivariant perverse sheaves on X_n that do not have composition factor IC_n . We will try to understand this subcategory and how much information it contains to understand $Perv_{G_n}(X_n, \mathbb{k})$.

Let $X_{1,1}$ be the principal open set in X_n with matrices whose first term $x_{1,1}$ is non-zero and

$$H := \left\{ \begin{pmatrix} x & 0 \\ v & A \end{pmatrix} \times \begin{pmatrix} y & w \\ 0 & B \end{pmatrix} \mid x, y \in \mathbb{C}^\times, v, w^T \in \mathbb{C}^{n-1} \text{ and } (A, B) \in G_{n-1} \right\}$$

be a subgroup in G_n . Observe that there is only one orbit i.e. X_n^n contained entirely in complement of $X_{1,1}$. Moreover, the subset $X_{1,1}$ is H -stable which gives the following:

Lemma 20: Let $j : X_{1,1} \hookrightarrow X_n$ be the inclusion map. Then

$$j^* : Perv_{G_n}(X_n, \mathbb{k})_n \hookrightarrow Perv_H(X_{1,1}, \mathbb{k}) \quad (5)$$

is fully faithful.

Proof. The exactness property comes from j is an inclusion of open set. Let $F, G \in \text{Perv}_{G_n}(X_n, \mathbb{k})_n$. Let i be the complementary closed embedding to j and consider the distinguished triangle,

$$i_! i^! G \rightarrow G \rightarrow j_* j^* G \rightarrow .$$

Applying the functor $\text{Hom}(F, -)$, we get

$$\text{Hom}(F, i_! i^! G) \rightarrow \text{Hom}(F, G) \rightarrow \text{Hom}(F, j_* j^* G) \rightarrow \text{Hom}(F, i_! i^! G[1]) \rightarrow .$$

Observe that ${}^p H^0(i^! G)$, ${}^p H^0(i^* F)$ are supported on complement of $X_{1,1}$. The only orbit contained entirely on complement of $X_{1,1}$ is X_n^n which implies that the only simple subobject it may contain is IC_n^n . Since $F, G \in \text{Perv}_{G_n}(X_n, \mathbb{k})_n$, we get

$$\begin{aligned} {}^p H^0(i^! G) &= 0 \\ {}^p H^0(i^* F) &= 0. \end{aligned}$$

This shows $i^! G \in {}^p D^{\geq 1}$ and $i^* F \in {}^p D^{\leq -1}$.

Using adjunction for i_* and i is closed embedding,

$$\text{Hom}(F, i_! i^! G[k]) = \text{Hom}(i^* F, i^! G[k]) = 0$$

for $k = 0, 1$. Hence,

$$\text{Hom}(F, G) \cong \text{Hom}(F, j_* j^* G)$$

Using adjunction on j_* i.e.

$$\text{Hom}(F, G) \cong \text{Hom}(F, j_* j^* G) \cong \text{Hom}(j^* F, j^* G),$$

we get our result. $\square$

Theorem 21: The category $\text{Perv}_{G_n}(X_n, \mathbb{k})_n$ is embedded in $\text{Perv}_{G_{n-1}}(X_{n-1}, \mathbb{k})$.

Proof. There is an embedding X_{n-1} in $X_{1,1}$ by sending $M \in X_{n-1}$ to $\begin{pmatrix} 1 & 0 \\ 0 & M \end{pmatrix} \in X_{1,1}$.

Set $G'_n := G_{n-1} \times \mathbb{C}^\times$ and embed it in H by sending $(A, B, c) \in G'_n$ to $\begin{pmatrix} c & 0 \\ 0 & A \end{pmatrix} \times \begin{pmatrix} c & 0 \\ 0 & B \end{pmatrix} \in H$. Since $\mathbb{C}^\times \subset H$ acts trivially on the copy of $X_{n-1} \subset X_{1,1}$, we get

$$\text{Perv}_{G_{n-1}}(X_{n-1}, \mathbb{k}) \cong \text{Perv}_{G'_n}(X_{n-1}, \mathbb{k}). \quad (6)$$

One can check, the natural map

$$\begin{aligned} \phi : H \times_{G'_n} X_{n-1} &\rightarrow X_{1,1} \\ (h, A) &\mapsto h \cdot A \end{aligned}$$

is well defined and shows

$$H \times_{G'_n} X_{n-1} \cong X_{1,1} \quad (7)$$

as H -variety. Using the inductive equivalence, j^* is fully faithful and the above isomorphism, we get an embedding of the category F as shown below

$$\begin{array}{ccccc}
Perv_{G_n}(X_n, \mathbb{k})_0 & \xrightarrow{5} & Perv_H(X_{1,1}, \mathbb{k}) & \xrightarrow{7} & Perv_H(H \times_{G'} X_{n-1}, \mathbb{k}) \\
& \searrow \text{embedding of category } F & & & \downarrow \text{Induction equivalence} \\
& & & & Perv_{G'}(X_{n-1}, \mathbb{k}) \\
& & & & \downarrow 6 \\
& & & & Perv_{G_{n-1}}(X_{n-1}, \mathbb{k})
\end{array}$$

□

Theorem 22: Let $X_n = M_n(\mathbb{C})$ and $G_n = GL_n(\mathbb{C}) \times GL_n(\mathbb{C})$. Then there is an equivalence of categories

$$Perv_{G_n}(X_n, \mathbb{k}) \cong \text{rep}(\widehat{AA_n})$$

for all $n \geq 1$.

Proof. We will prove this statement by induction on n . It is easy to check for $n = 1$. Assume $Perv_{G_{n-1}}(X_{n-1}, \mathbb{k}) \cong \text{rep}(\widehat{AA_{n-1}})$ for $n \geq 2$. The quiver associated to $Perv_{G_n}(X_n, \mathbb{k})_n$ is obtained by removing the n^{th} vertex from the quiver associated to $Perv_{G_n}(X_n, \mathbb{k})$ but keeping the same relations i.e.

$$0 \xrightleftharpoons[v_0]{u_0} 1 \xrightleftharpoons[v_1]{u_1} 2 \xrightleftharpoons[v_2]{u_2} \dots \xrightleftharpoons[v_{n-2}]{u_{n-2}} n-1.$$

The functor F constructed in 21 is fully faithful and the quiver associated to the categories $Perv_{G_n}(X_n, \mathbb{k})_n$ and $Perv_{G_{n-1}}(X_{n-1}, \mathbb{k})$ are same, we get

$$Perv_{G_n}(X_n, \mathbb{k})_n \cong Perv_{G_{n-1}}(X_{n-1}, \mathbb{k}).$$

We get the relations

$$u_i v_i = 0 = v_i u_i$$

for $0 \leq i \leq n-2$ and no other relations amongst them.

Since $\mathbb{T}$ is an equivalence of categories switching IC_0 to IC_n , the relations $u_0 v_0 = 0 = v_0 u_0$ forces the relation

$$u_{n-1} v_{n-1} = 0 = v_{n-1} u_{n-1}.$$

Thus, the quiver $Perv_{G_n}(X_n, \mathbb{k})$ has all the relations as the quiver $\widehat{AA_n}$

We still have to check if there are any relations with u_{n-1} and v_{n-1} with other u_i 's and v_i 's. The only possibility for such relations are of the form $u_{n-1} u_{n-2} \dots u_k = 0$ and $v_k v_{k+1} \dots v_{n-1} = 0$. In both the cases multiplicity of IC_k in the projective cover of IC_n is decreases to 0 which contradicts 17. Thus,

$$u_i v_i = 0 = v_i u_i$$

for $0 \leq i \leq n-1$ are all the relations in our quiver associated to $Perv_{G_n}(X_n, \mathbb{k})$. □

6 Applications

The functor $\mathbb{T}$ takes constant sheaf i.e. IC_0 to skyscraper sheaf i.e. IC_n but it is usually hard to find $\mathbb{T}(IC_k)$ for $0 < k < n$.

Let $S(k)$ be the simple representation in $\text{rep}(\widehat{AA_n})$ associated to IC_k in $\text{Perv}_{G_n}(X_n, \mathbb{k})$. Using our main result, we can try to understand indecomposable projective and injective objects from the quiver side. The projective cover of $S(n)$ is uniserial with composition factors $S(n), S(n-1), \dots, S(0)$. This implies that the projective cover of IC_n i.e. $\mathbb{T}({}^p H^0(j_{0!} \mathbb{k}_{X_n^0}[\dim X_n^0]))$ is uniserial with composition factor $IC_n, IC_{n-1}, \dots, IC_0$.

From 14, the perverse sheaf $\mathbb{T}({}^p H^0(j_{0!} \mathbb{k}_{X_n^0}[\dim X_n^0]))$ is uniserial with composition factor $\mathbb{T}(IC_0), \dots, \mathbb{T}(IC_n)$. The projective cover being uniserial and matching the composition factors, we get

$$\mathbb{T}(IC_k) \cong IC_{n-k}$$

.

6.1 Projective cover

We can also find the projective cover of IC_k understanding the properties of projective cover from quiver side.

Remark 23: In $\widehat{AA_n}$, the multiplicity of each simple $S(l)$ in projective cover of $S(k)$ is one.

The projective cover of $S(k)$ is the following representation

$$\mathbb{k} \xleftarrow{1} \mathbb{k} \xleftarrow{1} \cdots \mathbb{k} \xrightarrow{\cdots} \mathbb{k} \xrightarrow{1} \mathbb{k} \xrightarrow{1} \mathbb{k}$$

where the source is at vertex k .

For $0 \leq k \leq n$, let

$$j_0^k : \bigsqcup_{i=0}^k X_n^i \hookrightarrow X_n$$

be the open embedding of first $k+1$ strata and

$$j_k : X_n^k \hookrightarrow X_n$$

be the inclusion map of strata X_n^k in X_n .

Lemma 24: For $t < k$,

$$\begin{aligned} {}^p H^0(j_{0!}^k j_0^{k*} IC_t) &\cong IC_t \\ {}^p H^0(j_{0*}^k j_0^{k*} IC_t) &\cong IC_t \\ j_{0!*}^k(j_0^{k*} IC_t) &\cong IC_t. \end{aligned}$$

Proof. Let $i_0^k : \bigsqcup_{i=k+1}^n X_n^i \hookrightarrow X_n$ be the closed embedding. Using the following distinguished triangle,

$$j_{0!}^k j_0^{k*} IC_t \longrightarrow IC_t \longrightarrow i_{0*}^k i_0^{k*} IC_t,$$

we get the following long exact sequence

$${}^p H^{-1}(j_{0*}^k i_0^{k*} IC_t) \longrightarrow {}^p H^0(j_{0!}^k j_0^{k*} IC_t) \rightarrow IC_t \rightarrow {}^p H^0(i_{0*}^k i_0^{k*} IC_t) \rightarrow .$$

Using 3 and i_{0*}^k is t -exact,

$${}^p H^l(i_{0*}^k i_0^{k*} IC_t) = 0$$

for $l > -4$ which proves the first isomorphism. Applying Verdier dual to the first isomorphism, we get the second isomorphism. Using 3, 4, and lemma 3.3.34 in [Ach21] we get the third isomorphism. $\square$

Lemma 25: Let K be a perverse sheaf supported on X_n such that K does not have IC_k as a composition factor for $t \geq k$. Then,

$${}^p H^0(j_{0!}^k j_0^{k*} K) \cong j_{0!*}^k(j_0^{k*} K).$$

Proof. We will prove this using induction on number of composition factors in K . When K is simple i.e. $K \cong IC_t$ for $t < k$, we use lemma 24.

Suppose K has n composition factors. Then there exists an exact sequence

$$0 \rightarrow IC_t \rightarrow K \rightarrow K' \rightarrow 0$$

such that K' has $n - 1$ composition factors. Since K does not have IC_k as a composition factor, then so does K' . Thus, ${}^p H^0(j_{0!}^k j_0^{k*} K') \cong j_{0!*}^k(j_0^{k*} K')$. Using lemma 3.3.34 in [Ach21], we prove our lemma. $\square$

For $0 \leq k \leq n$, consider the perverse sheaves

$$\mathcal{P}_k := {}^p H^0(j_{0!}^k j_0^{k*} ({}^p H^0(j_{k*} \mathbb{K}[\dim X_n^k]))).$$

The next lemma gives the multiplicity of IC_t 's in $\mathcal{P}_k$'s.

Lemma 26: For $0 \leq k \leq n$ and $0 \leq t \leq n$, the multiplicity of IC_t in $\mathcal{P}_k$ is one.

Proof. Let K be the kernel of the map ${}^p H^0(j_{k*} \mathbb{K}[\dim X_n^k]) \rightarrow IC_k$. Applying the right exact functor ${}^p H^0(j_{0!}^k j_0^{k*} (-))$ to the exact sequence

$$0 \longrightarrow K \longrightarrow {}^p H^0(j_{k*} \mathbb{K}[\dim X_n^k]) \longrightarrow IC_k \longrightarrow 0,$$

we get a long exact sequence

$$\longrightarrow {}^p H^0(j_{0!}^k j_0^{k*} K) \longrightarrow \mathcal{P}_k \longrightarrow {}^p H^0(j_{0!}^k j_0^{k*} IC_k) \longrightarrow 0.$$

From 14 and 17, the kernel K does not contain IC_t for $t \geq k$, using 25 we also get exactness on left i.e.

$$0 \longrightarrow j_{0!*}^k j_0^{k*} K \longrightarrow {}^p H^0(j_{0!}^k j_0^{k*} ({}^p H^0(j_{k*} \mathbb{K}[\dim X_n^k]))) \longrightarrow {}^p H^0(j_{0!}^k j_0^{k*} IC_k) \longrightarrow 0$$

is an exact sequence.

Using 14 and 17, the kernel K has composition factors IC_t for $t < k$ with multiplicity one. Using 24, we get the multiplicity of IC_t for $t < k$ in $\mathcal{P}_k$ is one. Using 14, the object

$${}^p H^0(j_{0!}^k j_0^{k*} IC_k) \cong {}^p H^0(j_{k!} \mathbb{K}[\dim X_n^k])$$

has composition factors IC_t for $t \geq k$ with multiplicity one. Thus the multiplicity of IC_t for $0 \leq t \leq n$ in $\mathcal{P}_k$ is one. $\square$

Lemma 27: For $0 \leq k \leq n$ and $0 \leq t \leq n$,

$$\text{Hom}(\mathcal{P}_k, IC_t) = \begin{cases} \mathbb{k} & k = t, \\ 0 & \text{else} \end{cases}.$$

Proof. Observe that for $0 \leq t \leq n$, we have

$$\begin{aligned} \text{Hom}(\mathcal{P}_k, IC_t) &\cong \text{Hom}({}^p H^0(j_{0!}^k j_0^{k*} ({}^p H^0(j_{k*} \mathbb{k}[\dim X_n^k])), IC_t) \\ &\cong \text{Hom}({}^p \tau^{\geq 0}(j_{0!}^k j_0^{k*} ({}^p H^0(j_{k*} \mathbb{k}[\dim X_n^k])), IC_t) \\ &\cong \text{Hom}(j_{0!}^k j_0^{k*} ({}^p H^0(j_{k*} \mathbb{k}[\dim X_n^k])), IC_t) \\ &\cong \text{Hom}({}^p H^0(j_{k*} \mathbb{k}[\dim X_n^k]), j_{0*}^k j_0^{k*} IC_t) \\ &\cong \text{Hom}({}^p H^0(j_{k*} \mathbb{k}[\dim X_n^k]), {}^p H^0(j_{0*}^k j_0^{k*} IC_t)). \end{aligned}$$

For $t > k$, we have $j_{0!}^k j_0^{k*} IC_t = 0$ which implies

$$\text{Hom}(\mathcal{P}_k, IC_t) = 0$$

for $t > k$.

For $t < k$, we have ${}^p H^0(j_{0*}^k j_0^{k*} IC_t) = IC_t$. Using theorem 13, we get

$$\text{Hom}(\mathcal{P}_k, IC_t) = 0$$

for $t < k$

For $t = k$, we have

$$\begin{aligned} \text{Hom}(\mathcal{P}_k, IC_k) &\cong \text{Hom}({}^p H^0(j_{0!}^k j_0^{k*} ({}^p H^0(j_{k*} \mathbb{k}[\dim X_n^k])), IC_k) \\ &\cong \text{Hom}({}^p H^0(j_{k*} \mathbb{k}[\dim X_n^k]), {}^p H^0(j_{k*} \mathbb{k}[\dim X_n^k])) \\ &\cong \mathbb{k}. \end{aligned}$$

□

The following theorem proves $\mathcal{P}_k$ is the projective cover of IC_k .

Theorem 28: For $0 \leq k \leq n$, the perverse sheaf $\mathcal{P}_k$ is the projective cover of IC_k in $\text{Perv}_{G_n}(X_n, \mathbb{k})$.

Proof. Let P_k be the projective cover of IC_k in $\text{Perv}_{G_n}(X_n, \mathbb{k})$. From 27, the perverse sheaf $\mathcal{P}_k$ has a unique simple quotient IC_k which gives a surjective map from P_k to $\mathcal{P}_k$.

$$\begin{array}{ccc} P_k & \overset{\sim}{\dashrightarrow} & \mathcal{P}_k \\ & \searrow & \downarrow \\ & & IC_k \end{array}$$

Using 23 and 26, the multiplicity of simples matches in both perverse sheaves. The surjective map between perverse sheaves with same composition factors is an isomorphism which proves our theorem. □

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